

# STRATEGY SUBMISSION: TRACK A REPORT

Track A: Cross-Sectional Multi-Asset Dual-Factor Strategy | Inter IIT Tech Meet 15.0

## 1 Volatility Modeling & Risk-Parity Position Sizing

**Chosen Method:** Inverse Rolling Volatility Weighting ( $w_i \propto 1/\sigma_{i,30d}$ ) with gross leverage normalization.

**Technical & Economic Justification:** Asset return distributions exhibit kurtosis and volatility clustering. Markowitz Mean-Variance Optimization (MVO) or unconstrained GARCH-based covariance estimates suffer heavily from estimation error propagation.

- Estimation Robustness:** 30-day realized volatility provides a robust proxy isolating risk without matrix inversion.
- Equal Risk Contribution Proxy:** Inverse-volatility scaling balances risk across high- and low-beta assets.
- Turnover Mitigation:** Rolling windows avoid high-frequency parameter churn characteristic of daily GARCH models.

| Volatility Method      | Pros                         | Cons                                     | Selection Verdict |
|------------------------|------------------------------|------------------------------------------|-------------------|
| Sample Variance (MVO)  | Optimal under Normality      | Severe estimation error; extreme weights | Rejected          |
| GARCH(1,1) Dynamic     | Fast risk response           | High parameter churn & trading drag      | Rejected          |
| 30d Inverse-Vol Parity | Stable, robust, low turnover | Lags instantaneous 1-day spikes          | SELECTED          |

## 2 Cross-Sectional Ranking & Signal Evaluation

**Ranking Signal:** Dual-Factor Composite Score:  $S_{i,t} = 0.70 \cdot Z(R_{12m-1m,i,t}) + 0.30 \cdot Z(-1 \cdot R_{1m,i,t})$ .

**Economic Rationale:** *12m-1m Momentum* captures institutional underreaction to fundamental trends. Removing the recent 21 days ( $t - 21$  to  $t$ ) purges bid-ask bounce. *1m Short-Term Reversal* captures overreaction driven by liquidity provision bottlenecks.

### Statistical Testing & Correlation Analysis:

- Information Coefficient (IC):** Yields a mean IC of **+0.058** with a  $t$ -statistic of **3.42** ( $p < 0.0008$ ), confirming predictive power.
- Concentration & Correlation:** Pairwise correlation among top long assets ( $K = 5$ ) averages  $\rho = 0.28$ . Maximum weight cap of 20% ensures high diversification.

## 3 Portfolio Construction, Constraints & Rebalancing

**Optimization Choice:** Heuristic Risk Parity Top-K ranking selected over Quadratic Optimization to eliminate sample covariance noise.

**Rebalancing Logic & Frictions:** Periodic 5-day rebalancing matching alpha half-life. Execution at Open  $t + 1$  following Close signal at  $t$ . Frictions of **5 bps slippage** and **1 bp commission** deducted per trade.

- Long-Only Constraint:**  $w_{i,t} \geq 0 \ \forall i$ . Zero short positions.
- Gross Leverage Cap:**  $\sum_{i=1..K} w_{i,t} = 1.0$ . Strict full capital deployment.
- Single Asset Cap:**  $w_{i,t} \leq 0.25$ . Max position cap prevents single-asset overrun.

## 4 Out-of-Sample Validation, Sensitivity & Overfitting Analysis

**Walk-Forward Framework:** Evaluated across 1,484 trading days using backtest simulation.

### Actual Performance Metrics & Confidence Intervals:

| Key Metric                 | Point Estimate | 95% Confidence Interval / Detail            | Evaluation Status                     |
|----------------------------|----------------|---------------------------------------------|---------------------------------------|
| Total Return (%)           | 138.10%        | Initial: \$100,000.00 → Final: \$238,096.56 | 15.87% CAGR                           |
| Annualized Sharpe Ratio    | 0.81           | [0.11, 1.63]                                | Statistically Positive ( $p < 0.05$ ) |
| Sortino Ratio              | 0.92           | Downside Risk Normalized                    | Favorable Downside Asymmetry          |
| Max Drawdown (%)           | 38.20%         | Calmar Ratio: 0.42                          | Managed Structural Drawdown           |
| Trading Cadence & Turnover | 6,107 Trades   | 4.12 trades/day   Turnover: 19,684.65%      | 0 Leverage Violations                 |

**Parameter Sensitivity Analysis:** Over 85% of grid combinations yield Sharpe  $> 0.70$  across variations in factor weight, top-K, and rebalancing cadence.

**Overfitting & Deflated Sharpe Ratio (DSR):** Accounting for non-Gaussian skewness/kurtosis and  $N = 15$  trials,  $p_{DSR} = 0.018 < 0.05$ , confirming genuine factor alpha.